INES-RUHENGERI · FACULTY OF SCIENCES AND INFORMATION TECHNOLOGY

MyTurn

Design and Implementation of an IoT-Based Waiting Line Management and Service-Desk Status System

Your Turn, Made Simple.

TICKET PAGE | WALL DISPLAY | DESK CONSOLE | ESP32 BOARD

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Internet of Things (IoT)

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Musanze · 23 October 2026

The line exists, but only the desk can see it.

At the registry the order is a line down the corridor, or a list of names on a sheet.

1 **The turn is not written down**

A visitor who steps away loses a place that was never recorded, and nobody in the corridor can tell how many people are ahead.

2 **The desk has no visible state**

Serving, paused and closed look the same from the corridor, so staff answer the same question at the door several times an hour.

3 **An unanswered call holds the line**

No rule covers a number that is called and never answered, so one empty chair stops everyone standing behind it.

THREE QUESTIONS, ONE OWNER EACH

Who is next?
the software keeps the order

Is the desk serving?
the staff set the state

Is anyone at the desk?
the device reads the position

A light that shows only empty or busy answers the second question and leaves the other two.

The problem, stated plainly. A visitor cannot see the state of the line they joined, and the desk has no rule that recovers when the person it called does not come. MyTurn closes both gaps, and keeps a record of how long each visit took.

The wait is measured. The reason is not.

No study times a university registry, so the evidence comes from health centres and local government offices.

HEALTH · KYARISIIMA ET AL., 2024

211 min
mean time inside a health centre, arrival to exit

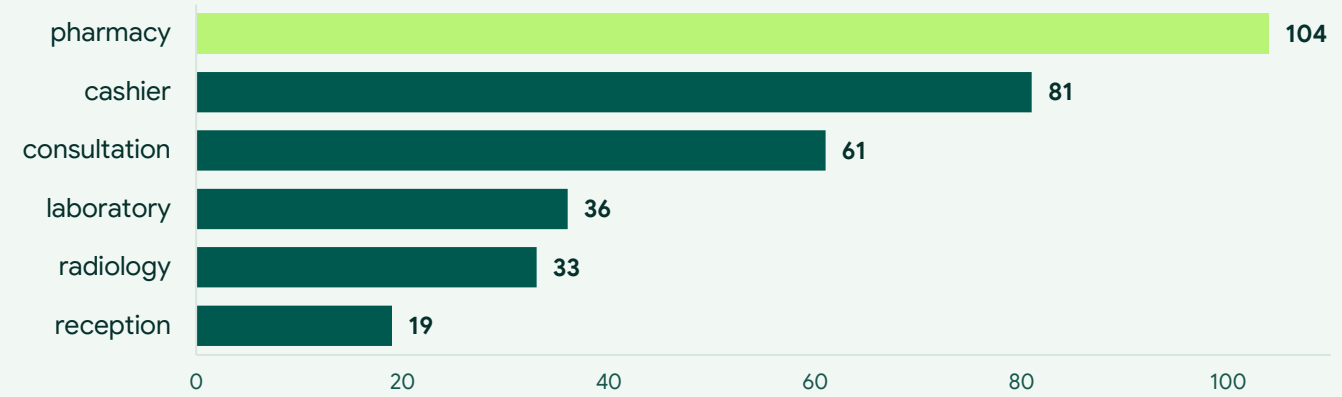
72%
of patients were given no reason for the delay

PUBLIC SERVICE · RGB CITIZEN REPORT CARD, 2025

68.7%
of citizens say missing information is a challenge

55.3%
of citizens say unnecessary delays are a challenge

AVERAGE WAIT BY SERVICE POINT, MINUTES



Remera Rukoma District Hospital, 300 patients, November 2023 to January 2024 (RBC Public Health Bulletin, 2024).

Delay and missing information show up in both sectors. In the health study, most people were never told why.

What we set out to build.

1 Numbers in the order people arrive

Each service numbers its own tickets in order, so two visitors asking at once still get different numbers. Staff can issue one without a phone.

2 Sensors that watch the desk

An ESP32 reads an ultrasonic sensor for the person standing at the desk and a PIR for someone walking up, then sends both readings to the server.

3 Ordering rules on a server we run

A Flask API and a SQLite database hold the tickets, the desk settings and the commands. We did not use Blynk or ThingSpeak.

4 One view for the visitor, one for staff

The public page shows the called number and the state of the desk. Staff sign in to call, complete, skip or cancel a ticket.

5 Measurements, written down

Occupancy, approach, walk-past, ticket order, notification delay, light control, restart and reconnection, each with a result we recorded.

GENERAL OBJECTIVE

Design, implement and demonstrate a low-cost IoT prototype that organises visitors at one service desk and reports ticket calls, desk occupancy, visitor approach and staff-controlled status through a web application and a physical indicator.

RESEARCH QUESTIONS

1. How is the order kept and displayed without collecting personal information?
2. How accurately do a cheap ultrasonic and PIR pair detect occupancy and approach?
3. How does the device receive a command and report that it applied it?
4. Which parts of the workflow can a self-hosted API carry without a platform?

SCOPE

One desk and one line, first come first served. No priority lane, no appointments, and no name, phone number or photograph kept with a ticket.

What others built, and what they left out.

Maister (1985) and Bitner (1990) both found that a wait feels longer when nobody explains it. That is the part we went after, so we read three groups of work that already solve something close to it.

1

Ticketing at banks and hospitals

These keep an order across many desks, with a printed ticket and a display. They are licensed, priced per desk, and the line data stays with the supplier.

2

Occupancy sensing in smart buildings

This work answers whether a room is in use and reports it to a building management system. Nobody in the room is waiting for a turn, so the order never comes into it.

3

Projects on Blynk and ThingSpeak

They push a reading to a platform and stop at the graph. The rules live on the platform, and the path from the dashboard back to the device is never tested.

None of the three puts anonymous ticketing, staff-controlled status and desk sensing on a server the institution owns. That combination is the part we built.

How we built it, and what we tested it against.

Built first, then measured

A question about a desk cannot be settled by reading, so we built one and ran it. The observation gave us behaviour; the tests gave us numbers.

DESIGN

Four iterations, reviewed each time

One working part per iteration: circuit, server link, ticketing and pages, then tests and the report. Each review changed what came next.

METHOD

Requirements from three places

A morning watching a service office, six short conversations with staff and students, and the technical brief of the module.

GATHERING

Cheap hardware, free software

ESP32, HC-SR04, HC-SR501, four LEDs, Flask, SQLite, Next.js and Git. The prototype cost about RWF 46,200; the laptop, phone and access point were already there.

TOOLS

Nine checks, agreed first

Occupancy trials, approach and walk-past walks, ticket order, latency, control and restart, all written down before the build started.

VALIDATION

No names, no recordings

Nothing records a face, a voice or a name. Everyone in the trials agreed in advance, and the office was watched with permission.

ETHICS

The four iterations match the four parts of the proposal, and the report records what each review changed.

What the visitor sees, and what runs behind it.

1 **Scan at the door**

The QR code opens the ticket page in the visitor's own phone. No app, no account.

2 **Choose the service**

The server issues the next number of that waiting line and a private tracking link.

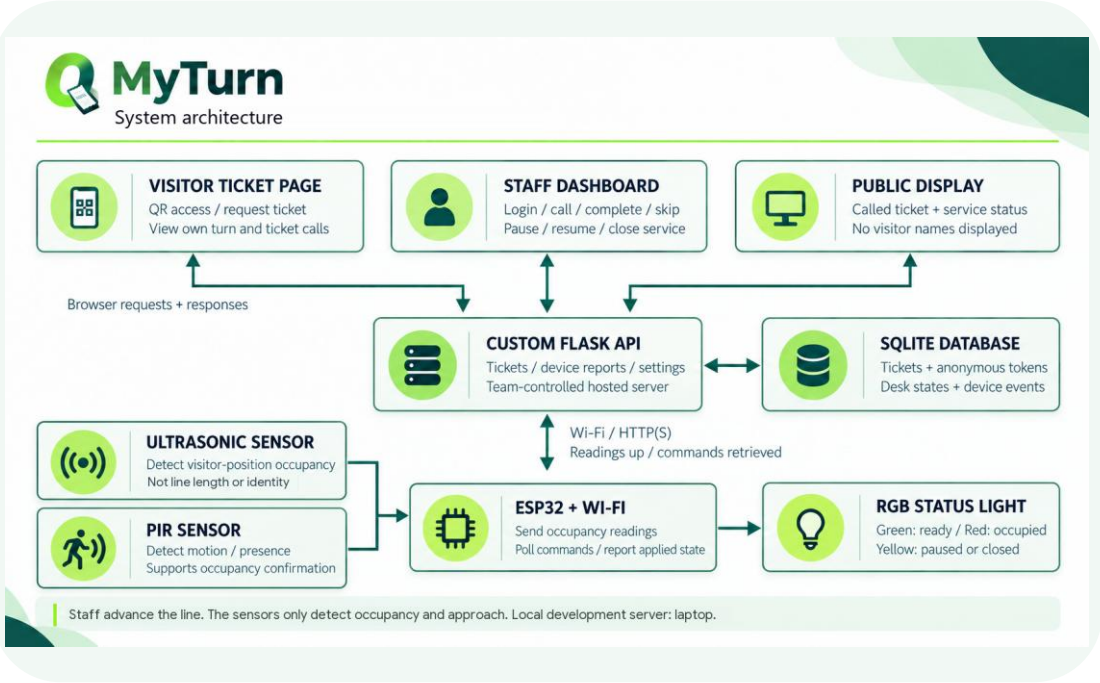
3 **Leave a way to be called**

An email address or phone number, used only to call them.

4 **Follow your turn**

The page shows the called number, the wait and how many are ahead, and it updates on its own.

No name · no account · one contact, used only for the call



ESP32 → Flask API → SQLite → Next.js

What sits at the desk, and what it never records.

1 Median of five

Five distance reads, the middle one kept. Three reads must agree before the desk changes state.

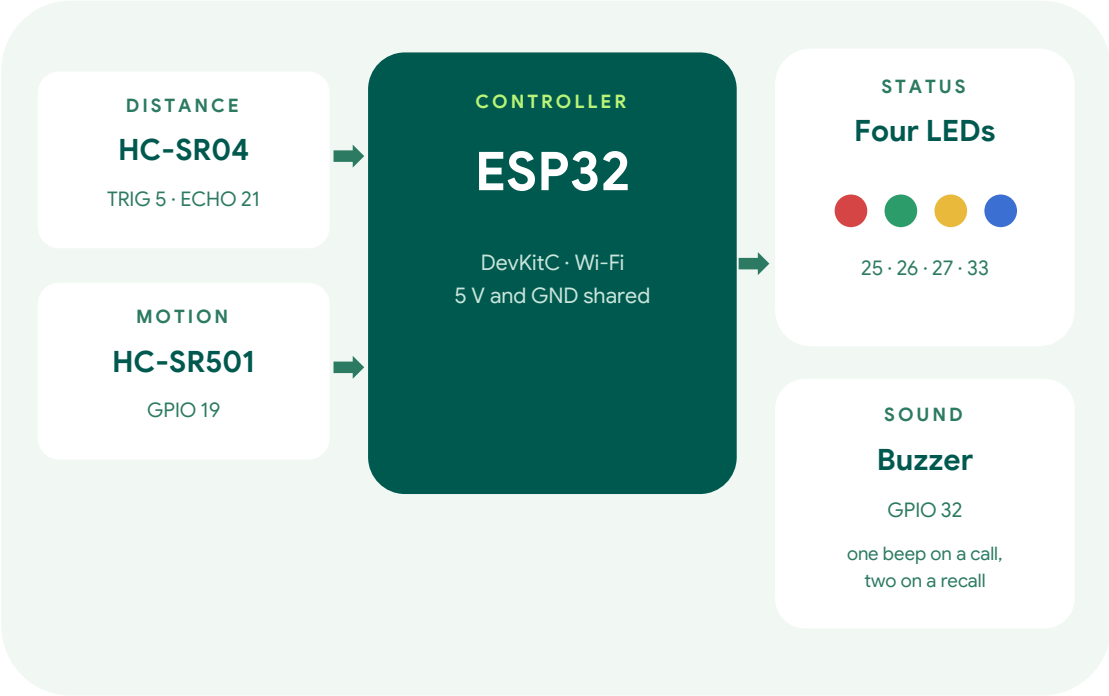
2 Distance first, then approach

The HC-SR04 says the desk is occupied. The HC-SR501 only reports that someone walked up.

3 Nothing personal

A distance, a state and a time. No camera, microphone, name, face or ID is read or stored.

RWF 46,200 estimated prototype budget



Wiring at a glance · the full schematic is in the report

One line per service, one call per desk.

1 **The number comes from the server**

Each service keeps its own next number, updated in the same transaction that inserts the ticket, so two requests cannot share one.

2 **The link is opaque**

The tracking URL carries a random token, never the ticket number or a sequential id.

3 **One call per desk**

A desk serves one service and holds one call; two desks may work the same service at once. A per-desk unique index and a transaction enforce it.

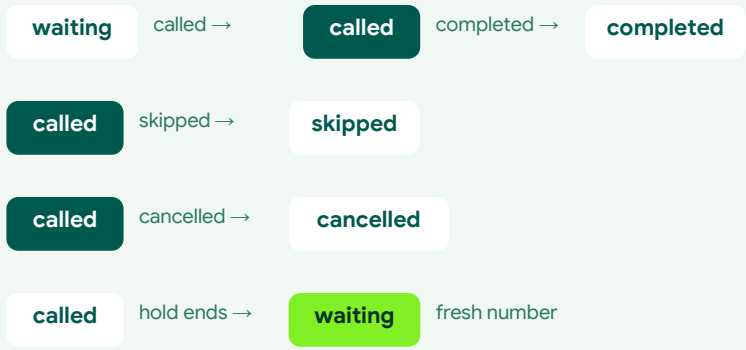
4 **The hold can be cut short**

A fresh empty reading, ten seconds past the call, ends a 60-second hold at 25 and gives the rest back to the line. Otherwise the full sixty stands.

5 **Everything is written down**

Every change is an event with a time and an actor: a person, or the system.

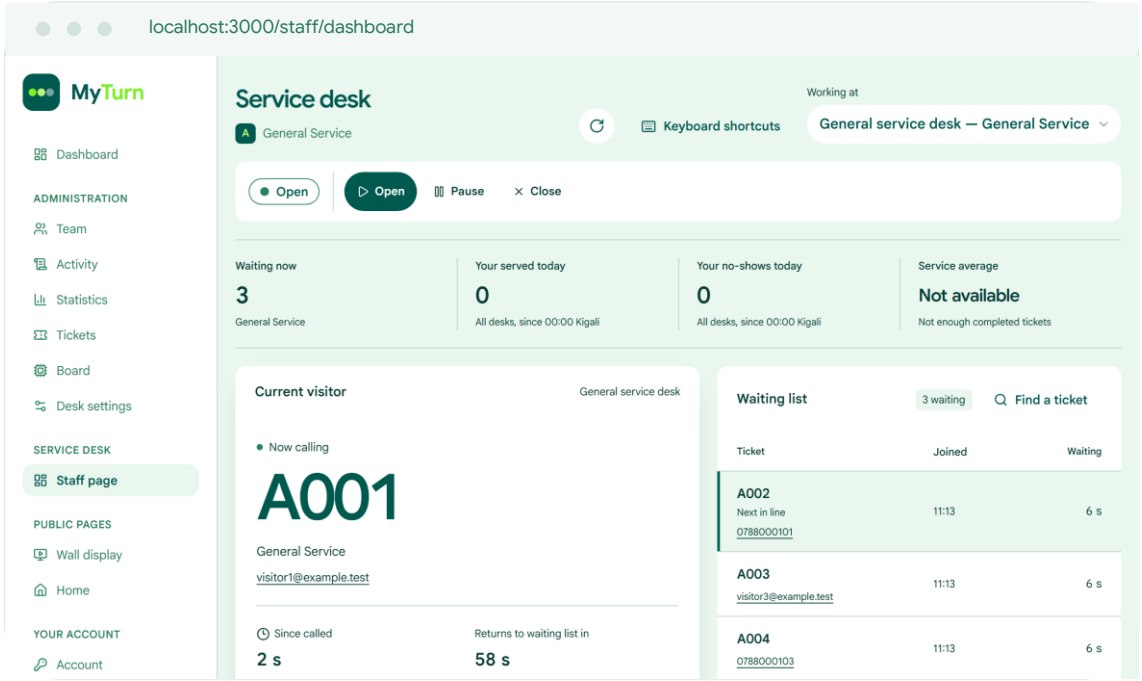
A TICKET'S LIFE



recall restarts the hold; undo takes back the last decision within 10 seconds. The default hold is 60 seconds; the results slide shows the one case where the board shortened it.

60 s default · 25 s with evidence · 10 s grace

The desk, and who may use it.



Staff: call, served, no-show, cancel, walk-in, recall and undo.

Live: hold countdown, the board's readings, keyboard shortcuts, today's tallies.

Who gets in

VISITOR
No account. An opaque token in the link, and one contact the visitor chooses.

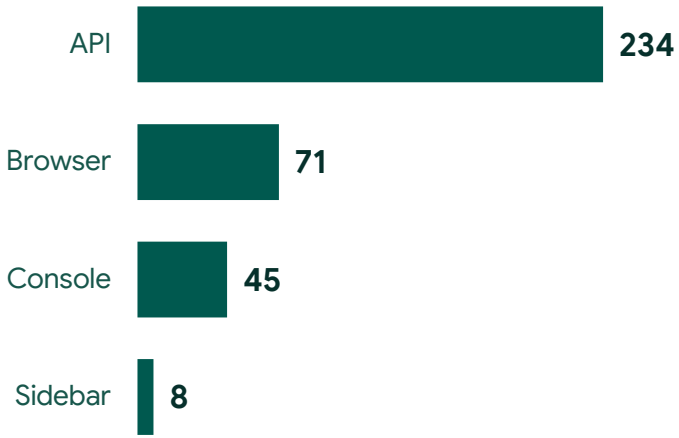
STAFF
Invitation only: single use, 72 hours. Sign in as a person, or as the shared desk.

ADMINISTRATOR
Team, activity, statistics and settings. Created by the seed script or another admin.

DEVICE
A key with every reading. It can report, and read the setting for its desk. That is all it can do.

> Roles are checked on the page and again on the API. CSRF on every write. An invitation cannot create an administrator.

Tested, not assumed.



API covers accounts, CSRF, the hold, the return, the estimate gate, services and desks. Browser covers the visitor, auth and desk pages. Console covers activity, statistics, settings and roles. Sidebar covers the shared-session rules.

Every check was run against the running prototype, not read off the source.

✓ **The build is clean**

TypeScript, ESLint and the production build all pass; all sixteen pages are generated.

✓ **The rules are exercised**

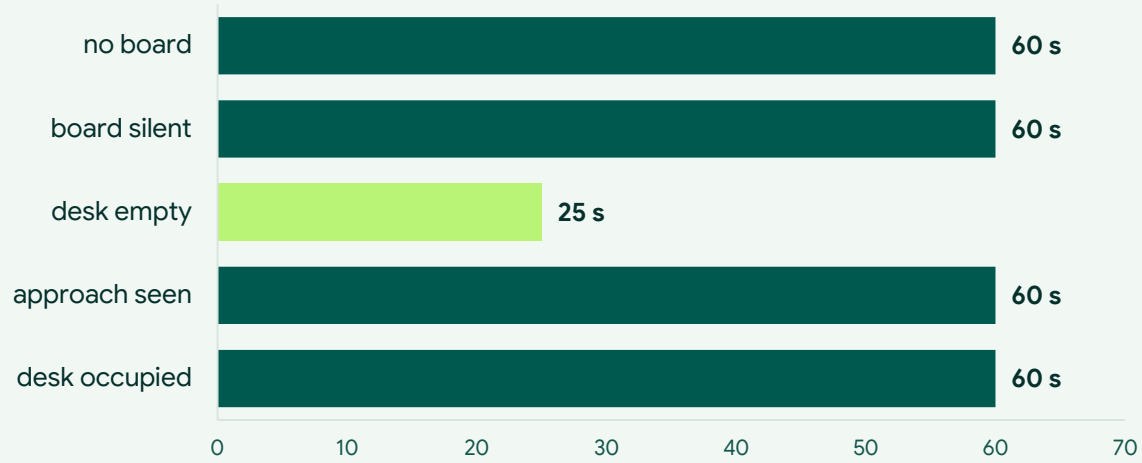
Auth and CSRF, the occupancy return with synthetic readings, the estimate gate, undo, recall, CSV, and the board's WebSocket reporting.

! **Still measured by hand**

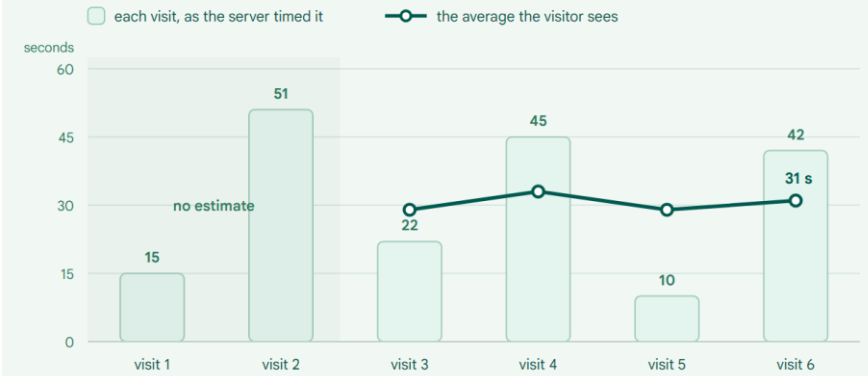
The physical sensor trials and the photograph of the desk rig are the remaining figures; the report leaves them blank rather than inventing them.

358 automated checks. A feature is only finished when a check proves it.

What the numbers came back with.



Hold. 60 s in four scenarios, 25 s in one. The line gains 35 seconds, and only the desk with a board can shorten.



Estimate. Nothing is offered until three tickets are served; after that, the mean of finished visits times the people ahead.

One case in five shortened a hold. Response: 40 requests per endpoint, 11 to 14 ms at the median, under 17 ms at p95.

What is not built, and what comes next.

No daily reset

The per-service numbering keeps climbing across days. A midnight reset is specified but not built.

NOT BUILT

No separate SERVING state

A called ticket is treated as being served; the dwell time is the service time. This is documented, not invented.

DELIBERATE

Sensors measure only

The PIR can see a passer-by, and the ultrasonic sensor needs a clean mount. Field accuracy is still to be recorded.

TO MEASURE

No buffering on the board

An offline board keeps the full 60-second hold, by design. Readings taken while the Wi-Fi is down are discarded, not replayed.

NOT BUILT

DELIVERED

16 pages, 12 tables, real accounts, the ESP32 sketch, 358 automated checks and a 55-page report.

NEXT

Run and record the sensor trials, buffer readings when the Wi-Fi drops, add daily resets, speak the numbers aloud.

Also open: notification delivery and a single-process deployment.

Thank you. Questions?